

6(a)	A – endodermal (cell) ; B – apoplast(ic) ;	A endodermis / passage cell	2
------	---	------------------------------------	----------

Question	Answer	Marks
6(b)	<i>five from:</i> 1 water potential gradient, between leaves and roots ; A water potential gradient across the root (root hair to xylem / across cortex) 2 <u>diffusion</u> out (via stomata) of <u>water vapour</u> ; 3 evaporation of water from mesophyll cell, surfaces / walls / membranes ; 4 (transpiration from leaves) creates transpiration pull ; 5 tension is, set up / present, in xylem vessels ; 6 cohesion between water molecules (<i>in xylem or in roots</i>) ; 7 hydrogen bonding between water molecules ; 8 adhesion of water molecules to <u>cellulose</u> (in wall of xylem vessels) ; A to hydrophilic parts of lignin 9 movement of water from apoplast through endodermis by osmosis ; 10 AVP ; e.g. <i>idea that</i> water columns extend from centre of root to root hairs (and to soil water)	5
6(c)	<i>two from:</i> <i>pathway C is slower accept ora for pathway B is faster</i> greater resistance ; water passes across membranes ; water flows through, cytoplasm / plasmodesmata ; A ora ref. to intercellular spaces A protoplast osmosis occurs ; R through plasmodesmata ref. to water passes through vacuoles ;	2
6(d)	<i>assume answer is about facilitated diffusion unless told otherwise – accept ora for active transport</i> <i>two from:</i> movement is, down a concentration gradient ; passive / ATP not required / (metabolic) energy not required ; carrier and channel proteins are involved ;	2